

# Laser induced breakdown spectroscopy online monitoring of laser cleaning quality on carbon fiber reinforced plastic

Wenju Wang<sup>a,b</sup>, Lanxiang Sun<sup>a,c,d,\*</sup>, Ying Lu<sup>a,c,d</sup>, Lifeng Qi<sup>a,c,d</sup>, Wei Wang<sup>a,c,d</sup>, Hongchao Qiao<sup>a,c,d</sup>

<sup>a</sup> State Key Laboratory of Robotics, Shenyang Institute of Automation, Chinese Academy of Sciences, Shenyang 110016, China

<sup>b</sup> Northeastern University, Shenyang 110006, China

<sup>c</sup> Key Laboratory of Networked Control Systems, Chinese Academy of Sciences, Shenyang 110016, China

<sup>d</sup> Institutes for Robotics and Intelligent Manufacturing, Chinese Academy of Sciences, Shenyang 110169, China

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## ABSTRACT

Carbon fiber reinforced polymer (CFRP) has attracted extensive attention in the industrial field due to its advantages of light weight and high hardness. The use of laser to clean the epoxy resin film on the surface of the carbon fiber is beneficial to improve the bonding performance. Combined with laser-induced breakdown spectroscopy (LIBS) technology, an on-line monitoring system for laser cleaning is designed to monitor the quality of laser cleaning in real time. The laser used in the experiment is a fiber laser widely used in laser cleaning, which has high cleaning efficiency, flexible use, and can be processed in a multi-dimensional space. The system is coupled with the scanning galvanometer which can be cleaned dynamically in all directions. The function of fast, real-time and high-resolution on-line monitoring of cleaning quality is realized. In this paper, prior information such as element composition and LIBS spectrum of epoxy resin and substrate carbon fiber in CFRP materials were obtained, and the differences were compared and analyzed. The results shown that the LIBS technology was feasible to monitor the cleaning quality. Subsequently, the experiment tested the influence of the laser process parameters on the LIBS characteristic spectrum, and it was confirmed that the average spectral intensity value at the wavelength (588.819 nm) was positively correlated with the average laser power. After that, according to the measured LIBS characteristic strength value, three evaluation grades of cleaning quality were effectively delineated. The piecewise linear fitting method was used to solve the value range of LIBS mapped by different grades, and the key data such as the optimal cleaning parameter threshold were calculated. Finally, scanning electron microscope (SEM) was used to observe the morphological characteristics of samples after laser cleaning under different grades, confirming that the LIBS characteristic line could effectively characterize the quality of laser cleaning.

## 1. Introduction

Carbon fiber reinforced polymer (CFRP) has been widely used in aircraft manufacturing, automobile, structural reinforcement engineering, new energy development, leisure products and many other fields due to its advantages of light weight and strong hardness [1–3]. In the production process, it is necessary to use adhesive to connect multiple CFRPs to enhance the physical properties of materials, which is called bonding treatment. In order to achieve the ideal bonding effect, the epoxy coating on the surface of CFRP needs to be removed in advance. With the development of laser cleaning technology, laser cleaning has

entered various fields of industry in a novel way, such as laser decontamination [4], laser descaling [5], welding pretreatment [6], oxide layer removal [7], etc. Laser cleaning technology is a new type of surface pollutant removal technology, which has the advantages of environmental protection, high precision, non-contact, etc. Therefore, laser cleaning technology is a kind of green cleaning technology with development potential [8–12]. However, if the laser irradiation time is too long, it will cause excessive damage to the material substrates. On the contrary, if the irradiation time is too short, it will lead to incomplete cleaning of the surface of the material. In order to meet the cleaning requirements without causing damage to the substrate, a feasible non-

\* Corresponding author at: No.114 Nanta Street, Shenhe District, Shenyang City, 110000, China.

E-mail address: [sunlanxiang@sia.cn](mailto:sunlanxiang@sia.cn) (L. Sun).

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destructive online monitoring technology is urgently needed. At present, the main monitoring methods include surface imaging and measuring process performance parameters, but these methods are not real-time and have some limitations.

Laser-induced breakdown spectroscopy is an approximate non-destructive measurement technique, which has little restriction on the cleaning objects. It has been used in many researches such as monitoring the laser cleaning of cultural relics and metals, etc. [13,14,16]. This technology can effectively avoid damage to the substrate material and has real-time performance, which makes it possible to realize automatic optimization control in the laser cleaning process. Asmus et al. [17] applied laser-induced breakdown spectroscopy to underwater archaeology and performed laser cleaning of historical relics under water. LIBS monitoring technology could effectively avoid rapid oxidation of underwater archaeological objects and reduce the degree of damage. Fotakis et al. [18] studied the use of LIBS technology to monitor the process of laser cleaning of limestone in ancient buildings. Stefan Kleina et al. [19] used LIBS technology to analyze the peak intensity of characteristic spectral lines in the study of laser cleaning of medieval glass and sandstone to determine whether the laser cleaning was over. ENEA et al. [20] used LIBS as a diagnostic tool for online monitoring and cleaning control of ancient marble in the Mediterranean area. Staicu A et al. [15] used visible light and ultraviolet laser pulses to conduct LIBS studies on different natural varnishes that were normally or artificially aged. Chen, et al. [21] measured the spectral signal of plasma in the process of paint removal based on LIBS technology and studied the change of spectral characteristic peak intensity of elements in paint with time. Tong, Y. Q., et al. [22] in the study of laser cleaning of copper coins with surface contamination, according to the fact that the continuous spectral lines in the spectrum of copper coins disappeared and only copper spectral lines existed, it indicated that the samples have been cleaned. Hai, R., et al. [23] proposed a dual-pulse laser-induced breakdown spectroscopy technique based on the vertical reheating of the thin layer of impurities on the surface of the first mirror of the Tokamak. This work combined laser cleaning with LIBS online analysis technology, which could accurately and quickly remove the co-deposited impurity layer without damaging the optical surface of the first mirror.

Although LIBS technology has made some achievements in monitoring the laser cleaning process, there are few researches on online monitoring and cleaning of carbon fiber composite materials. Furthermore, most of the lasers used in the research of LIBS to monitor the cleaning quality are KrF-excimer lasers or Q-switched Nd:YAG laser. The cleaning speed is slow and the cleaning processing space is limited. Secondly, the method of judging whether the cleaning is completed can be based on the continuous change of the LIBS spectrum (vanishing of peaks and generating of new element peaks) in the cleaning process, which will affect the efficiency of laser cleaning and increase the difficulty of real-time monitoring of laser cleaning process.

This article made innovations on above basis, the cleaning system used fiber pulse laser (JPT) which was coupled with the scanning galvanometer and could be cleaned dynamically in all directions. The system realized the function of fast, real-time and high-resolution online monitoring of the cleaning quality.

In our previous studies, the Na(I) atomic lines at the wavelengths of 588.819 nm and 589.411 nm in the LIBS spectrum could characterize the cleaning quality of carbon fiber composite materials [24]. In this paper, the element composition information of different structural layers of CFRP materials was obtained by using X-ray energy spectrum analyzer (EDS) and LIBS technology. The process of laser cleaning epoxy resin and the principle of monitoring cleaning quality by LIBS were analyzed. Subsequently, the influence of laser processing parameters on LIBS characteristic spectral lines was tested, and the average spectral intensity at the wavelength (588.819 nm) was positively correlated with the average laser power. After that, the piecewise linear fitting method was used to analyze the relationship model between the LIBS characteristic spectrum and the average laser power. The optimal threshold of

laser cleaning average power was determined to be 8.2w, and the LIBS characteristic spectrum value mapped by this parameter threshold was 128.22 a.u. Finally, scanning electron microscope (SEM) was used to observe the morphological characteristics of samples under different cleaning grades, confirming that the LIBS characteristic spectrum could effectively characterize the laser cleaning quality.

## 2. Experimental part

### 2.1. Laser cleaning and monitoring system

The laser cleaning monitoring system mainly includes fiber laser, dichroic mirror, scanning galvanometer, quartz lens, control system, spectrum acquisition system, mobile station, etc. The structure diagram of the system is shown in Fig. 1.

The fiber laser emits a laser beam and is incident on a dichroic mirror that is horizontally deflected at an azimuth angle of 45°. The laser beam reflected by the dichroic mirror incident on the scanning galvanometer. Under the control of related software, the scanning galvanometer oscillates according to the set path, so as to realize the process of laser beam cleaning on the plane.

Plasma is produced when laser acts on the thin layer of sample surface. The light emitted by the laser-induced plasma passes through the dichroic mirror in the opposite way. Under the convergence of quartz lens, the light transmits to the entrance slit of spectrometer through optical fiber. The spectrometer converts the collected optical signals into spectral data and stores them on the computer.

The output wavelength of the fiber laser is 1064 nm. The maximum output power is 20w and the pulse width is 100 ns. The model of the spectrometer in the system is AvaSpec-ULS2048L. The spectrometer has 2048 pixels CMOS array detector. The resolution of the spectrometer is 0.15 nm. The wavelength coverage is from 453.17 nm to 696.05 nm. The integration time of the spectrometer can be adjusted from 30  $\mu$ s to 50 s.

When the test platform works, the software of PC control terminal sets the cleaning parameters (laser output parameters, cleaning path planning, etc.) and gives the cleaning command which will send out the start signal of laser cleaning. The collection trigger signal of the spectrometer is connected with the output signal of the laser so as to ensure that the spectrometer is turned on to collect the plasma spectrum in real time.

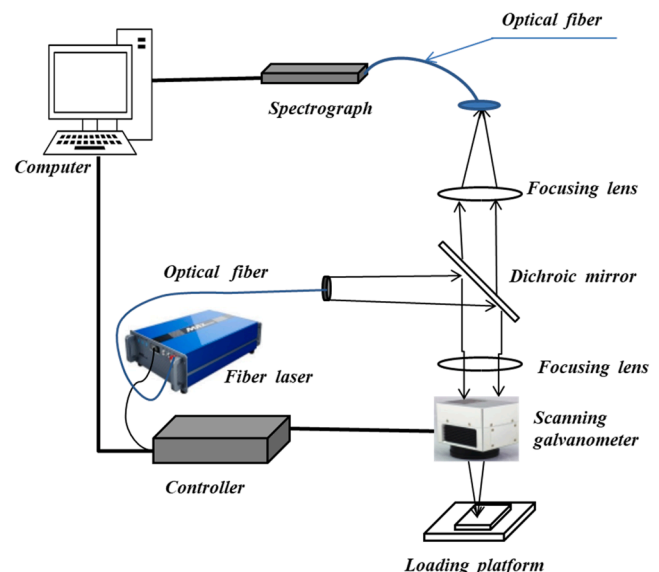


Fig. 1. Laser cleaning and monitoring system structure.

## 2.2. Experimental materials

The samples used in the experiment are carbon fiber reinforced polymer materials. The epoxy resin of the film layer is evenly distributed and the structure has no cracks and irregular shapes. The surface of the sample film is smooth and has good reflectivity. The specifications of the material are 15 cm × 10 cm × 4 mm. Scanning electron microscope (SEM) was used to observe and analyze the morphology of the substrate and the film layer. The image obtained is shown in Fig. 2.

Fig. 2 can clearly observe that the material carbon fiber substrate and the epoxy resin film layer are clearly distinguished in terms of morphology. The carbon fiber substrate of the material presents uniform spacing of fine stripes, but the surface epoxy resin film layer is relatively smooth. In the experiment, some areas are selected from the surface epoxy resin and carbon fiber substrate, labeled 605 and 606, respectively. For different labeled areas, an X-ray energy spectrum analyzer (EDS) is used to obtain the composition information of the film epoxy resin and carbon fiber substrate elements as shown in Fig. 3 and Fig. 4.

According to the test report chart of EDS, it can be seen clearly that the carbon fiber material used in the experiment contains four main elements (C, H, O, S) in both the substrate and the film layer.

## 3. Results and discussion

### 3.1. LIBS monitoring and analysis

In the laser cleaning process, the mechanism of laser cleaning the surface of composite materials is a complex physical and chemical process. Firstly, the carbon fiber composite is irradiated by laser beam, which absorbs heat continuously, melts and sublimates gradually to complete the physical ablation process. As the surface energy of the sample continues to accumulate, the vaporized gas around the material will undergo a reheating process, and the internal covalent bonds of the gas molecules will break, which completes the chemical ablation process. At the same time, the ionization of atoms in the material generates a large number of free electrons and forms a plasma. Plasma decays and emits element specific radiation, which can be detected by the spectrometer. For the carbon fiber composite material, driven by the inherent dynamics of plasma expansion and externally applied air flow, the evaporated epoxy resin material is partially removed from the interaction area, and finally a crater is formed on the surface of the sample, that is, the laser cleaning process of epoxy resin is completed. Under the action of the laser spot, the diameter of the laser outer circle spot can be observed. Because the laser beam profile is approximately Gaussian, the temperature at the center of the laser beam is higher, and the cleaning depth of the inner ring is relatively deeper. The result is shown in Fig. 5.

The spectrum emitted by the plasma plume depends on the elemental composition of the ablated material. Before laser cleaning, it is necessary

to obtain a priori information of material standard spectral wavelength. The laser frequency is set to 25 kHz and the integration time of the spectrometer is set to 50 ms. The cleaning speed set to 1000 mm/s and the scan line length set to 20 mm. That is, the laser turn-on time is 20 ms. Under this parameter, the spectrum which accumulates about 500 laser plasmas is obtained. The pulse laser is applied to the carbon fiber composite material sample and the LIBS reference spectrum is collected as shown in Fig. 6.

Fig. 6 shows that the spectrum contains a very high continuous background. In the integration time of 50 ms, the spectrum of 500 plasma life cycles is collected by the spectrometer. In the early stage of plasma generation, the continuous background spectrum is dominant due to the interaction of blackbody radiation, bremsstrahlung radiation and compound radiation. With the disappearance of the plasma, the continuous background spectrum decreases rapidly and the discrete line spectrum representing the characteristics of the atom appears gradually. Due to the high frequency of the fiber laser, the spectrum collected by the spectrometer is the cumulative spectrum of the plasma life cycle so that the spectrum contains a very high continuous background. The reference spectrum (a) of the carbon fiber substrate contains clear characteristic spectral lines of the elements. The strong spectral line can be indicated by a digital sign, and it can be clearly observed that there are two characteristic peaks in the substrate, which are located at 588.819 nm and 589.411 nm respectively. Compared with the wavelength of standard spectral line, each spectral line can represent the determined element, as shown in Table 1. The Na I state in the table represents the atomic line of Na. The small deviation between the experimental wavelength and the standard wavelength may be caused by the wavelength calibration error or the relatively low wavelength resolution of the spectrometer. No effective characteristic spectral lines of elements are observed in the film layer reference spectrum (b). Therefore, the Na (I) lines at 588.819 nm and 589.411 nm can be used to characterize the cleaning quality of carbon fiber composites. All non-metal elements (C, O, N, S) in carbon fiber composite materials are not easy to excite plasma and the characteristic lines of these elements are not within the effective detection range of the spectrometer, so the spectrometer cannot detect the effective information of the relevant elements. Due to the limited resolution of the EDS equipment, trace amounts of Na(I) atomic information cannot be detected. The experiment uses a combination of LIBS and EDS equipment to analyze the main elements and trace elements of the composite material.

### 3.2. Laser parameter test

The laser used in the experiment is a fiber laser with a maximum average power of 20W. With the aid of the scanning galvanometer, the laser beam can achieve large-scale area cleaning. When the laser emits a laser beam, the scanning galvanometer converts the light spot into a linear arrangement according to the given path in the software system. If the spot reaches the end point of the scanning line, the scanning galvanometer will deflect along the vertical direction, and the new area will be cleaned. The path of fiber laser cleaning is downward scanning in turn, as shown in Fig. 7.

In order to ensure that the carbon fiber surface after laser removal of epoxy resin has a good cleaning quality, it is necessary to select appropriate laser cleaning parameters. According to research and analysis, the main influencing factors of cleaning quality are laser parameters and spatial parameters. Laser parameters mainly include laser pulse width, power, frequency, etc., and spatial parameters mainly include scanning speed and spot size. The adjustment range of each parameter and its influence on the cleaning effect are shown in Table 2.

The average laser power parameter has great influence on the laser cleaning effect. The factors of frequency, scanning speed, and spot size are superimposed on each other, which indirectly affect the laser energy density. The experimental pulse width is set to 100 ns, and the scan line spacing  $D = 0.1$  mm.

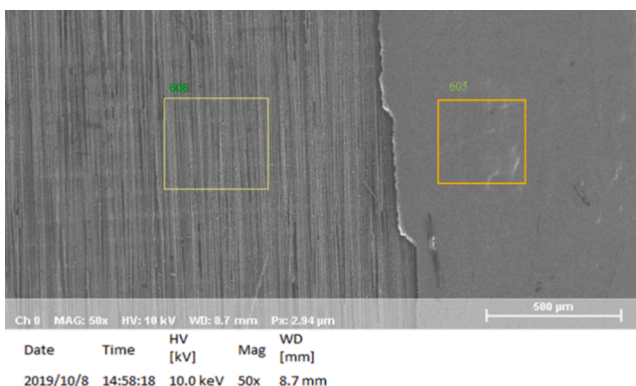


Fig. 2. Morphological characteristics of the substrate and the film layer.

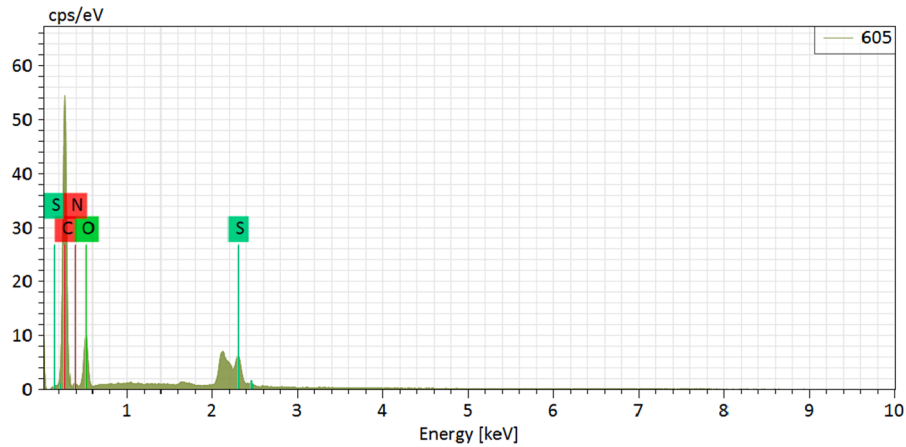


Fig. 3. Elemental composition analysis of epoxy resin.

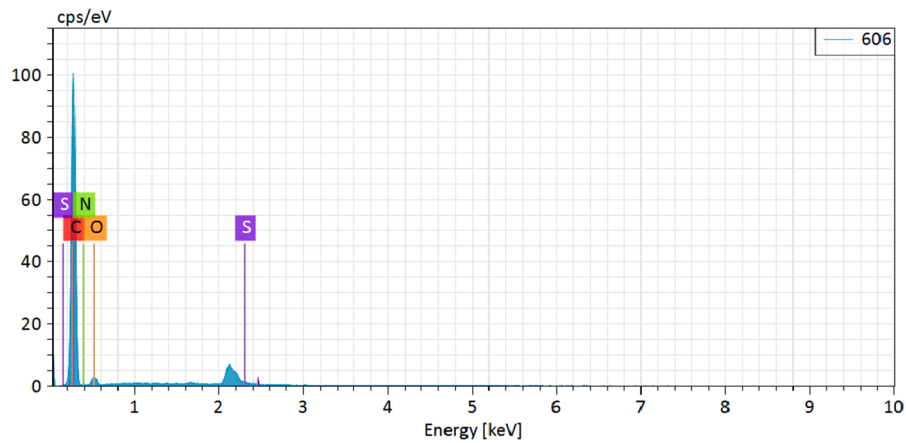


Fig. 4. Elemental composition analysis of substrate carbon fiber.

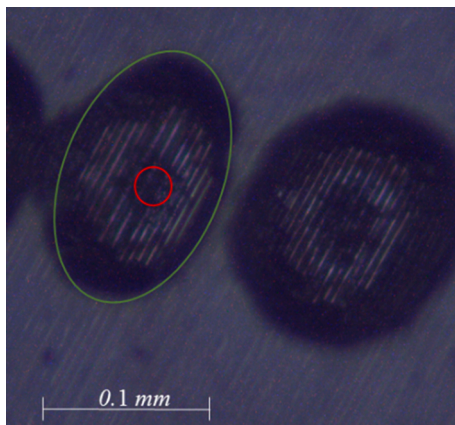


Fig. 5. The morphology analysis diagram of carbon fiber material under the action of laser spot.

The size of the whole area of the experimental laser cleaning CFRP is 6 mm × 7 mm, that is, the experimental horizontal line scanning cleaning length is 6 mm, and the longitudinal cleaning width of the area is 7 mm. Due to the longitudinal spacing distance of the laser spot array is 0.1 mm, the scanning galvanometer will be deflected 70 times in the cleaning area. With each longitudinal swing of the scanning galvanometer, the cleaning position area will change and the spectrometer will simultaneously collect a piece of spectral information. The

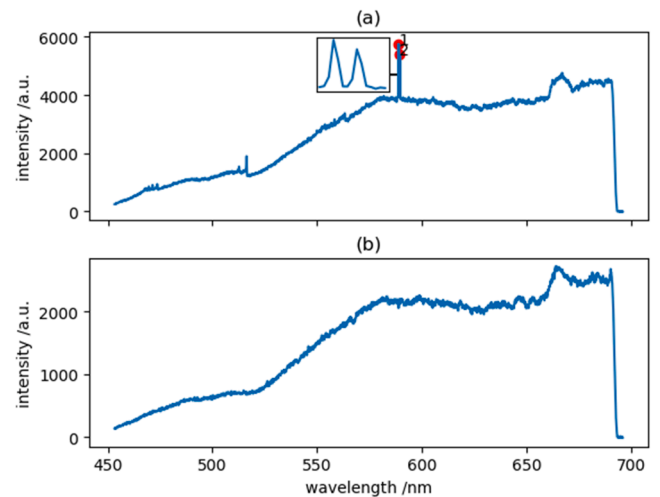


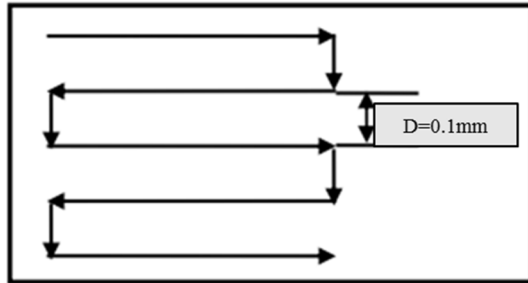
Fig. 6. Reference spectrum collected by spectrometer. Subgraph (a) shows the reference spectrum of carbon fiber substrate and subgraph (b) shows the reference spectrum of film layer.

spectrometer will collect 70 spectral data to reflect the overall cleaning quality in the target cleaning area. LIBS spectral data with different average power parameters is collected at 25 kHz pulse frequency and the characteristic peak intensity of wavelength at 588.819 nm is extracted. According to the characteristic intensity information of the element, the



**Table 1**  
Element characteristic lines.

| Label | Atomic spectrum | Standard-wavelength/(nm) | Experimental-wavelength/(nm) | Wavelength-deviation/(nm) |
|-------|-----------------|--------------------------|------------------------------|---------------------------|
| 1     | Na I            | 588.819                  | 588.955                      | 0.136                     |
| 2     | Na I            | 589.592                  | 589.414                      | 0.177                     |



**Fig. 7.** Scanning galvanometer motion track.

**Table 2**  
Selection and setting of cleaning process parameters.

| Laser parameters | Power/(w) | Pulse width/(nm) | Frequency/(kHz)     | Sweep speed/(mm/s) | Spot size/(mm) |
|------------------|-----------|------------------|---------------------|--------------------|----------------|
| Value            | 6–13      | 100              | 25–85               | 1000               | 0.1            |
| Effect           | Great     | Little           | Joint determination |                    |                |

relationship between the laser parameters and the characteristic peak spectral line intensity is analyzed, as shown in Fig. 8.

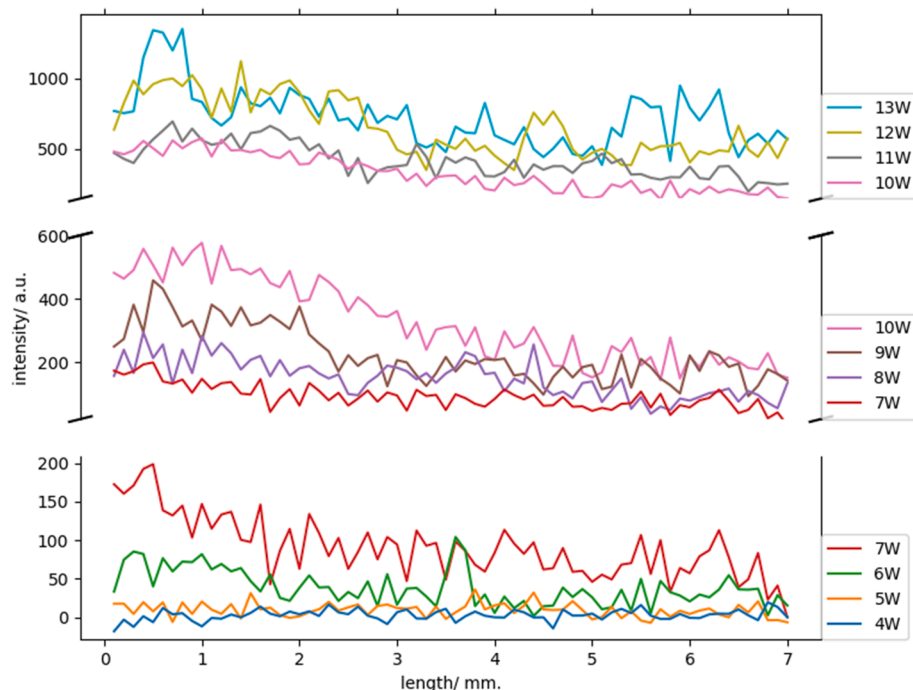
As the longitudinal deflection angle of the scanning galvanometer is constantly changing, the cleaning position will have a longitudinal distance difference. The greater the longitudinal deflection angle of the galvanometer, the weaker the laser beam spot power density, and the

lower the intensity value of the LIBS characteristic spectrum collected by the spectrometer. In addition, the thickness of the epoxy film layer on the CFRP surface is not completely uniform. The thicker the epoxy film is, the less likely it is to excite the plasma of the substrate characteristic information elements, so the intensity of the characteristic peaks detected by the spectrometer is weaker. In the cleaning area with the same laser average power and different distances, the spectral intensity value of the characteristic wavelength (588.819 nm) fluctuates, but it is stable in a certain range. That is, the characteristic peak intensity of the substrate element at the wavelength of 588.819 nm is generally positively correlated with the average laser power, which confirms the feasibility of the characteristic LIBS intensity value to monitor the quality of laser cleaning.

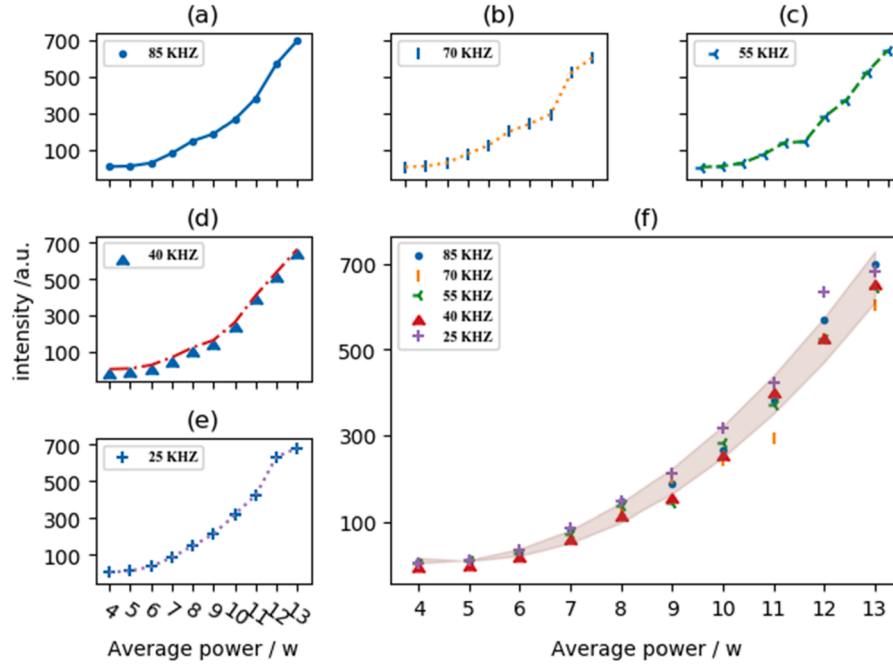
After that, the experiment set the range of pulse frequency from 25 kHz to 85 kHz and the frequency is taken every 15 kHz interval, so there are 5 groups of tests. The laser average power ranges from 4w to 13w and one set of values is taken every 1w interval, so there are 10 sets of tests in total. By combining different groups of parameters in turn, 50 groups of experiments can be obtained.

In order to better evaluate the quality of the overall area of laser cleaning, the average spectrum is introduced. Average spectrum can overcome the problems of poor repeatability of collected spectrum data and ensure the accuracy of data. The 70 spectra collected in the target cleaning area are averaged once. Experimental analysis of the average spectral intensity under different parameter combinations, as shown in Fig. 9.

Fig. 9(f) shows that when the average laser power is the same, the change of pulse frequency has little effect on the spectral line intensity of characteristic elements, so the experiment can ignore the influence of pulse frequency. Fig. 9(a) ~ (e) shows that the intensity of the average spectrum at the characteristic wavelength (588.819 nm) increases in turn with the increase of the average laser power under the same frequency. According to the data value of the spectral characteristic peak intensity, the cleaning quality grade can be delineated reasonably and effectively. When the average laser power is low, the plasma of the carbon fiber substrate cannot be excited and the LIBS spectral line intensity value is low. This shows that the laser ablation fails to act on the carbon fiber substrate and cannot completely remove the surface film,



**Fig. 8.** LIBS intensity curve in the cleaning area under different average laser power.



**Fig. 9.** The curves of the average spectral intensity change under different parameter combinations (a) - (e) the relationship between the line intensity and the average laser power at different frequencies (f) Contrast diagram of spectral line strength at different frequencies.

which belongs to the uncleaned grade. When the average laser power is too high, the intensity of the substrate feature information collected by the spectrometer is more intense. This shows that the laser has penetrated the substrate material and the ablation depth is too deep, which is an excessive damage grade. When the average laser power is selected appropriately, it is the critical or optimal grade of laser cleaning quality. Under different cleaning grades, the change trend of the spectral data is obviously different, so it is particularly important to analyze the threshold of the average laser power based on the spectral data.

### 3.3. Calculation of laser average power threshold

In this section, the correlation model between the average laser power and the characteristic peak intensity of LIBS can be established based on the experimental observation data. Usually, the mathematical model should be robust. It can overcome the influence of fluctuation and has good expected effect. In order to better evaluate the model, the evaluation indicator is the coefficient of determination ( $R^2$ ), which is calculated by Formula (1).

$$R^2 = 1 - \frac{\sum_{i=1}^m (Y_i - \hat{Y}_i)^2}{\sum_{i=1}^m (Y_i - \bar{Y})^2} \quad (1)$$

In Formula (1),  $Y_i$  represents the predicted value of the  $i$ -th sample data point,  $\hat{Y}_i$  represents the true value of the  $i$ -th sample,  $\bar{Y}$  represents the average of all sample data, and  $m$  represents the sample size.

#### 3.3.1. Piecewise linear fitting

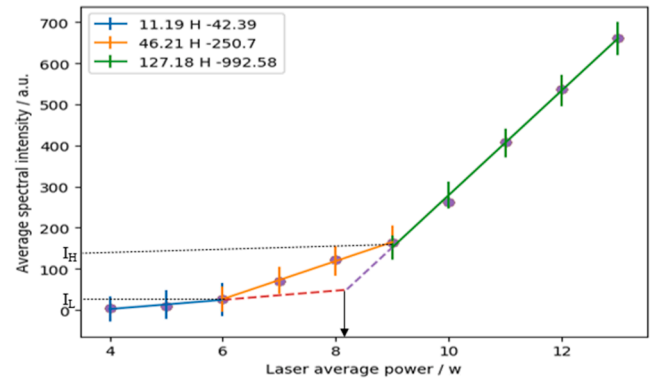
Because there are 3 grades of laser cleaning surface morphology, the average laser power is divided into 3 intervals. The average power value range of each interval is assumed to be  $[4, a]$ ,  $[a, b]$ ,  $[b, 13]$ , corresponding to different cleaning quality grades in turn. The experiment enumerates all possible intervals, and uses the piecewise linear fitting method to select the interval division with the largest  $R^2$ , namely  $a = 6$ ,  $b = 9$ . In different laser power parameter intervals, we analyze the relationship between the characteristic spectral line intensity and laser average power. The characteristic spectral line intensity fitting curve is

shown in Fig. 10. The fitting equation of the characteristic spectral line intensity is shown in Formula (2).

$$I = \max(11.19H - 42.39, 46.21H - 250.70, 127.18H - 992.58) \quad (2)$$

In Formula (2),  $I$  represents the peak intensity of the characteristic element, and  $H$  represents the average laser power.

Fig. 10 shows that the three cleaning quality grades are mapped to their respective intervals in turn and the steepness of the changes has significant differences. The intersection of the two dotted lines is the optimal threshold for the average power of laser cleaning. According to calculation and analysis, the optimal threshold of laser cleaning average power is 8.2w and the LIBS characteristic peak threshold ( $\tilde{I}$ ) mapped by this threshold is 128.22 a.u. In the experiment, the LIBS characteristic peak threshold ( $\tilde{I}$ ) of the substrate is used as the measurement baseline. The cleaning quality is the best when the measured LIBS characteristic peak fluctuates around the baseline. In the critical grade of cleaning quality, the acceptable range of laser average power is 6 ~ 9w and the range of LIBS characteristic peak value (26.56 a.u. ~ 165.19 a.u.) fitted by the model is recorded as ( $I_L \sim I_H$ ). When the characteristic peak of LIBS collected by the spectrometer is higher than  $I_H$  or lower than  $I_L$ , it



**Fig. 10.** Piecewise linear fitting graph between average power and the characteristic peak intensity.

indicates that the cleaning quality is unqualified so that the system will reinitialize the laser parameters. In the actual cleaning and processing, the acceptability degree of the cleaning quality of the material is different, and the reasonable range of the LIBS values mapped by the critical grade is effectively adjusted according to the actual needs.

### 3.3.2. Analysis of surface morphology

50 groups of parameters were selected to test the cleaning quality of each group. After cleaning, the morphology of the carbon fiber composite surface is shown in Fig. 11 (a). A series of complex physical and chemical processes occur when fiber laser irradiates carbon fiber sheets, including thermal behaviors such as pyrolysis, ablation, and oxidation. The epoxy resin film layer on the surface of the carbon fiber absorbs the laser input heat flow energy, and then oxidizes, decomposes and vaporizes to produce strong white smoke. Fig. 11(a) shows that when the laser power continues to increase from low to high, the epoxy film layer will gradually peel off from the material, and then the surface film layer has signs of ablation and is gradually destroyed. If the laser power is too high, the material film layer has obvious black spots, indicating that the substrate has been exposed to the air. That is, the substrate has significant damage.

When the laser is cleaning the target area for each set of parameters, the scanning galvanometer is deflected 70 times longitudinally and the spectrometer collects 70 pieces of spectral information. A total of 3,500 spectral data graphs are collected for 50 sets of parameters. When the measured LIBS characteristic peak is higher than  $I_H$ , it is classified as excessive damage grade. Similarly, the measured LIBS characteristic peak is lower than  $I_L$ , which is classified as uncleaned grade. Fig. 11(b) shows LIBS online monitoring cleaning quality grade maps in sequence to the target cleaning area in Fig. 11(a) under different parameter combinations.

The study uses scanning electron microscope (SEM-sigma 300) to observe the morphological features of the material surface after laser cleaning. According to different microscopic characteristics, the difference of cleaning morphology under different LIBS intervals is compared. The laser cleaning sample areas under three grades are extracted and the measured LIBS characteristic data are significantly different at different

grades. Cut each group of samples into 5 mm \* 5 mm squares and then take out the samples after spraying gold for 10 min. Then implement the surface dust removal work, and finally put the sample in the scanning electron microscope object storage and start to observe the surface topography. The magnification should be set to 50 $\times$ , 100 $\times$ , 200 $\times$ , 300 $\times$ , 500 $\times$ .

The intensity of the LIBS characteristic spectrum collected by the spectrometer is higher than  $I_H$  under this condition of average power parameter of 10w, which is excessive damage grade. The epoxy resin film on the surface of the carbon fiber is completely peeled off and is burned through to varying degrees and there are small holes in the morphology. The fiber diameter becomes larger and carbon fiber is obviously broken and has been carbonized. Using SEM respectively magnify 50 $\times$ , 200 $\times$ , 300 $\times$ , 500 $\times$  times to observe the morphological characteristics, as shown in Fig. 12.

The intensity of the LIBS characteristic spectrum floats near the measurement baseline ( $\bar{I}$ ) under this condition of average power parameter of 8w, which is critical grade. The LIBS peak value of the characteristic element fitted by the model is 118.98 a.u. The epoxy resin on the surface is completely peeled off without any damage and the substrate is not obviously broken. Therefore, it shows the accuracy of LIBS monitoring the cleaning quality. Using SEM respectively magnify 300 $\times$ , 500 $\times$  times to observe the morphological characteristics, as shown in Fig. 13.

The intensity of the LIBS characteristic spectrum is lower than  $I_L$  under this condition of average power parameter of 6w, which is uncleaned grade. Most of the energy of the laser is absorbed by the impurities and epoxy on the surface. The low power density of the laser can only remove part of the area and cannot completely remove the epoxy film layer. At the same time, the remaining epoxy resin film layer also absorbs the laser energy and has a major change in chemical composition and structure. The bonding force is weakened between the epoxy resin film layer and the carbon fiber. Using SEM respectively magnify 300 $\times$ , 500 $\times$  times to observe the morphological characteristics, as shown in Fig. 14.

Combining the cleaning characteristics under the above different parameters show that the laser cleaning quality can be characterized by

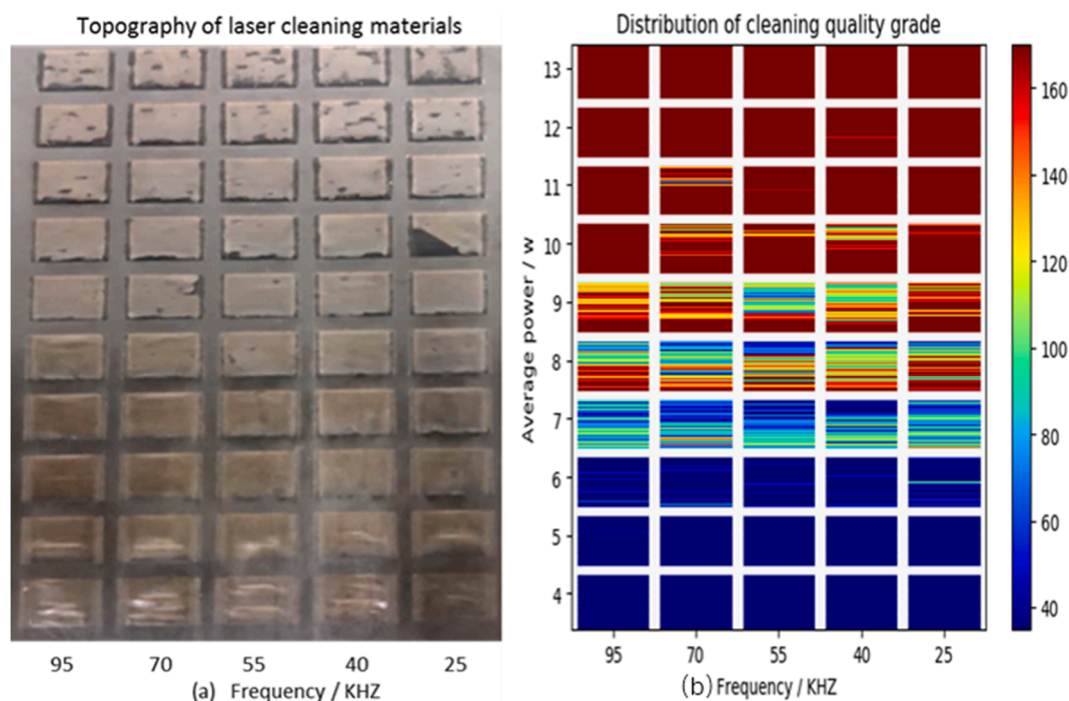


Fig. 11. Morphology and grade analysis results of samples after cleaning; (a) Topography of sample in target area after laser cleaning; (b) Distribution of cleaning quality grades in target area.



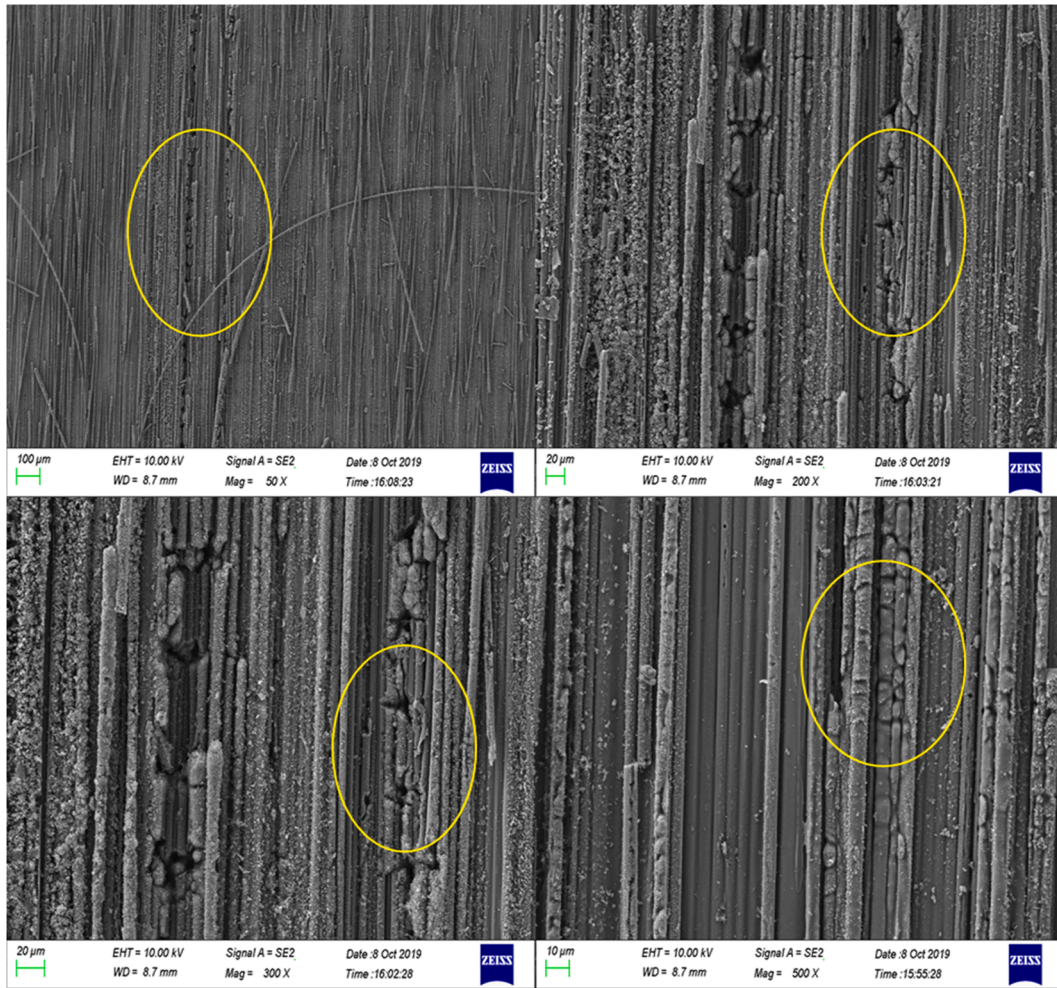


Fig. 12. Topography analysis of different magnification (50  $\times$ , 200  $\times$ , 300  $\times$ , 500 $\times$ ) after laser cleaning under excessive damage grade.

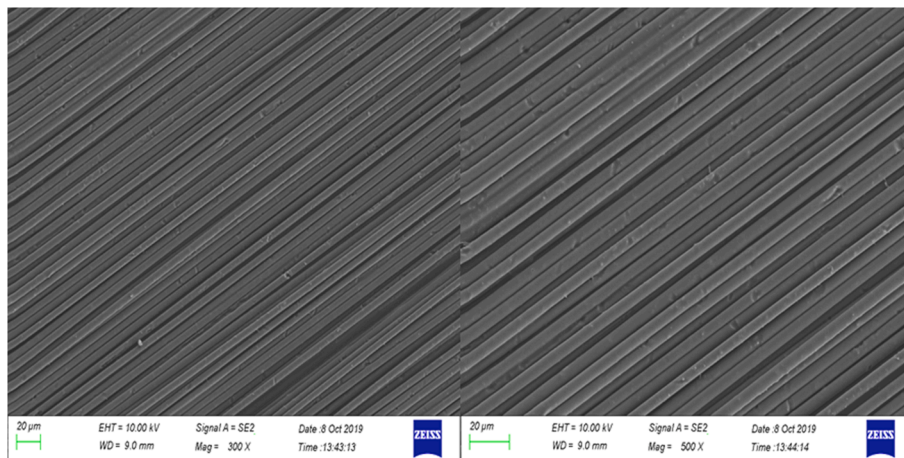


Fig. 13. Morphology of the sample material is amplified by (300  $\times$ , 500 $\times$ ) times after laser cleaning under critical grade.

the spectral line intensity of LIBS collected by spectrometer. In the process of laser cleaning automatic monitoring, the laser power output value should be adjusted in real time so that the LIBS measured value should be maintained near the baseline. Because of the thickness of the epoxy resin film layer is not exactly the same, only fixing the process parameter thresholds cannot achieve better cleaning quality. If the spectrometer detects that the measured characteristic peak value is

much higher than the baseline, the system should reduce the laser power to avoid excessive damage to the substrate again. In the same way, if the measured peak value is much lower than the baseline, it indicates that the material film is not peeled off enough so that the laser power should be increased. In the later stage, according to the change of the measured LIBS characteristic peak value, negative feedback adjustment will be realized and the laser output parameters will be automatically adjusted



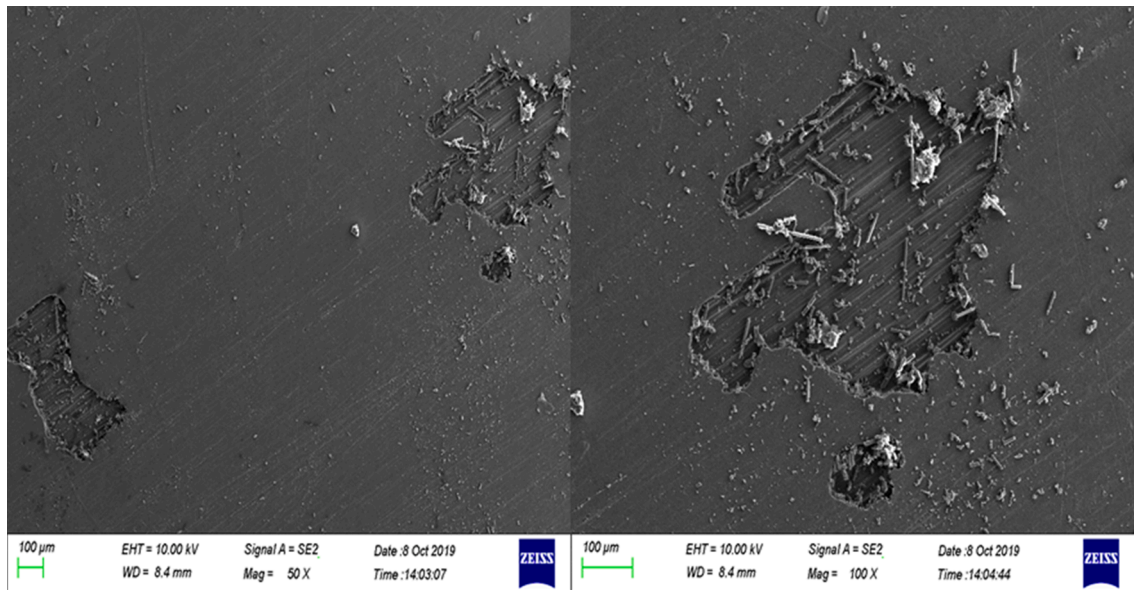


Fig. 14. Morphology of the sample material is amplified by (300 ×, 500×) times after laser cleaning under uncleaned grade.

to realize automatic optimization control.

#### 4. Conclusion

The article combines laser-induced breakdown spectroscopy (LIBS) technology to design an online monitoring system to monitor the quality of laser cleaning in real time. The laser used in the experiment is a fiber laser widely used in laser cleaning, which has high cleaning efficiency, flexible use, and can be processed in a multi-dimensional space. Firstly, the prior information such as element composition and LIBS spectrum of epoxy resin and carbon fiber substrate is obtained. The results show that both of them contain four main elements (C, H, O, S) and the substrate carbon fiber contains Na element but the epoxy resin does not. Comparing and analyzing the difference of LIBS spectra of different substances, the Na atomic line at wavelength of 588.819 nm in LIBS spectra can characterize the cleaning quality. Subsequently, the experiment tests the influence of the laser parameters on the LIBS characteristic spectrum, confirming that the average spectral intensity at the wavelength (588.819 nm) is positively correlated with the average laser power. After that, the piecewise linear fitting method is used to analyze the relationship model between the LIBS characteristic spectrum and the average laser power. The optimal threshold of laser cleaning average power is determined to be 8.2W and the LIBS characteristic spectrum value mapped by this parameter threshold is 128.22 a.u. According to the actual measured LIBS characteristic strength value, three evaluation grades of cleaning quality are effectively delineated. If the measured LIBS characteristic peak fluctuates near the baseline, the cleaning quality is the best. Finally, the SEM is used to observe the micro-topography characteristics of different grades. The LIBS characteristic spectrum which can characterized the cleaning quality is confirmed.

#### Author contributions

Wenju Wang mainly takes charge of writing and experiments, Lanxiang Sun is the supervisor of Wenju Wang, and plays a guiding role in this paper.

#### Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence

the work reported in this paper.

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#### Appendix A. Supplementary material

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.optlastec.2021.107481>.

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